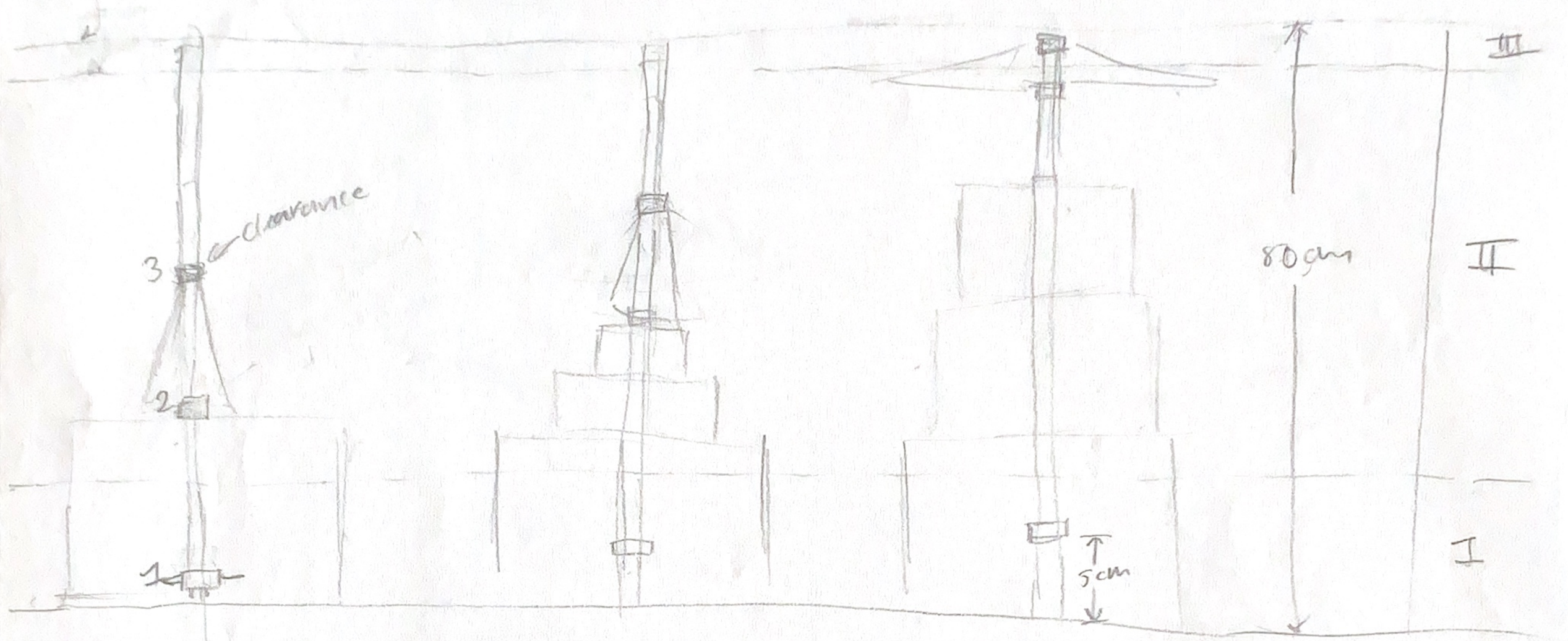
UMBRELLA

I Lead Screw

Umbrella opens after whole free is expanded



All ϕ2cm (M20)

Zone I: very slow
Zone II: fast
Zone III: slow

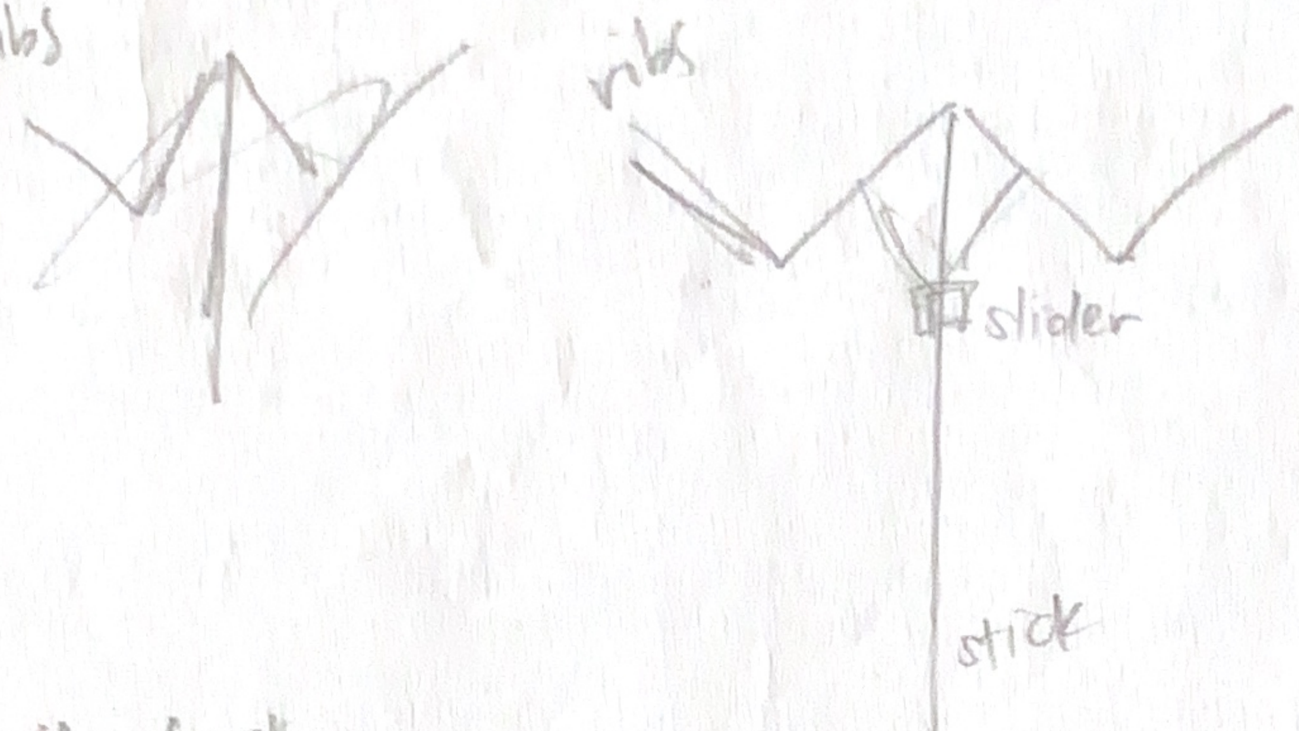
Formula for speed: lead = pitch x # starts

I: Very slow: 4mm pitch, 1 start

II: fast: 8mm pitch, 4 start

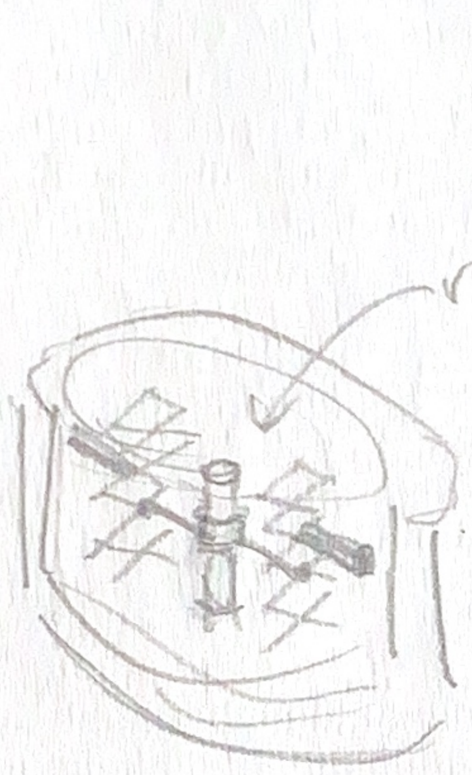
1/21/26

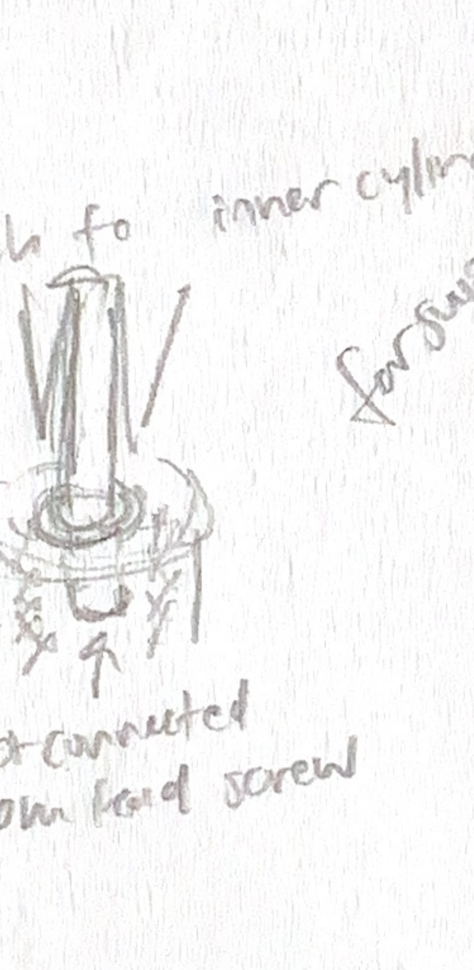
Components of an umbrella:

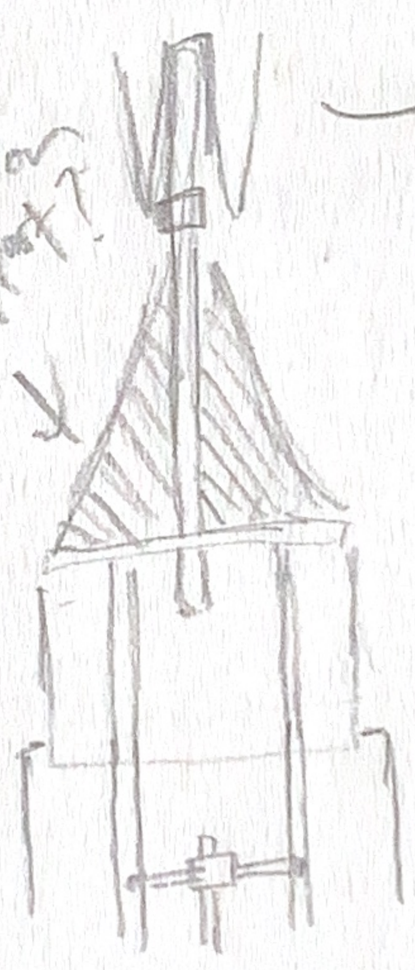


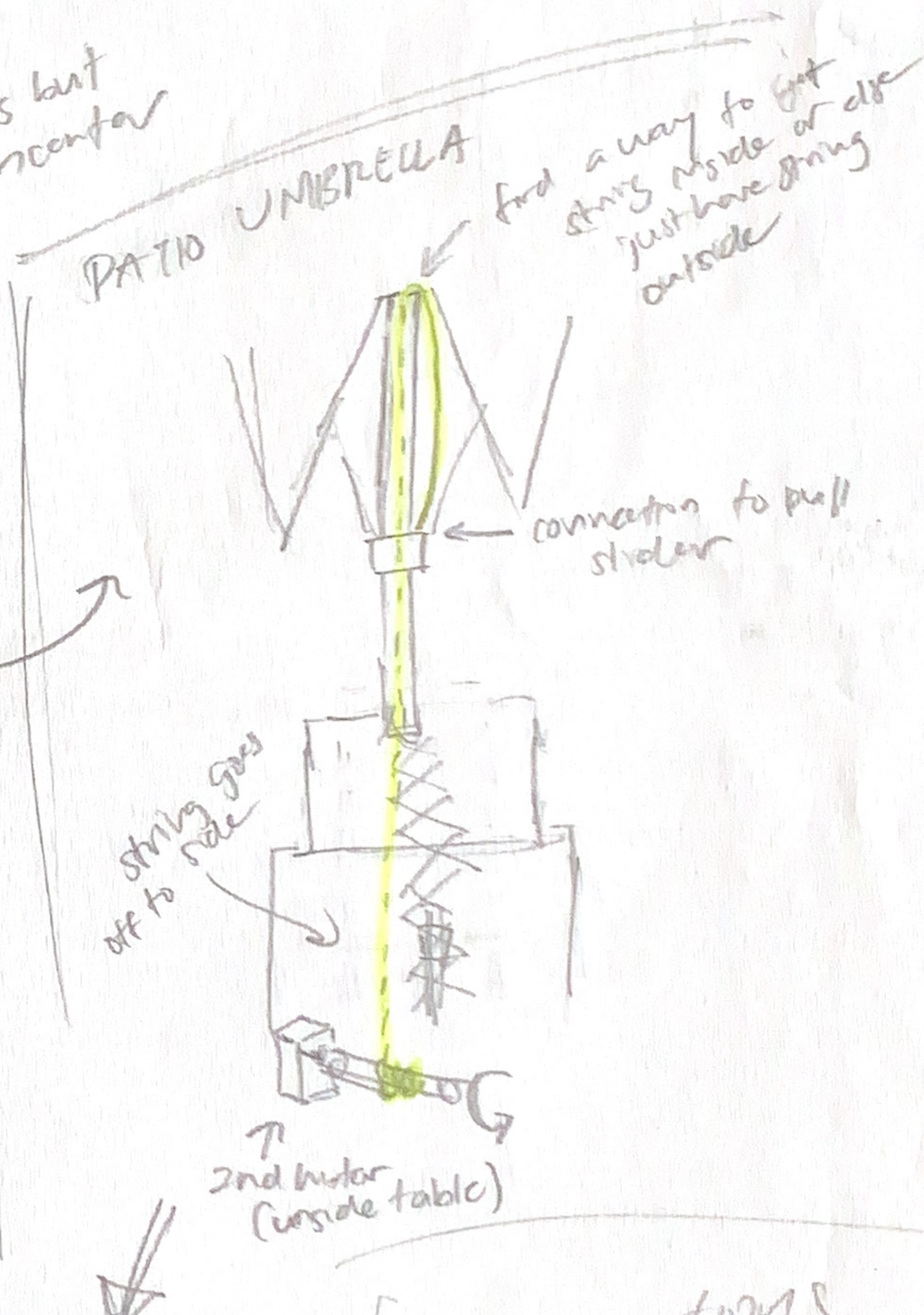
- unit of slider should be controlled
- stick should remain stationary relative to slider

the frame:









- patio umbrella motor 2 starts moving once the scissor is done

or starts at same time, but goes very slowly

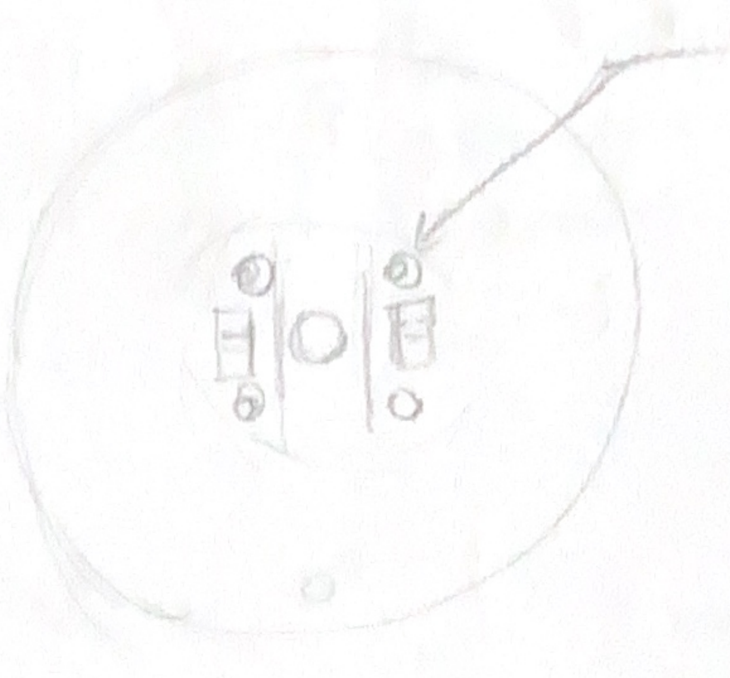
Questions

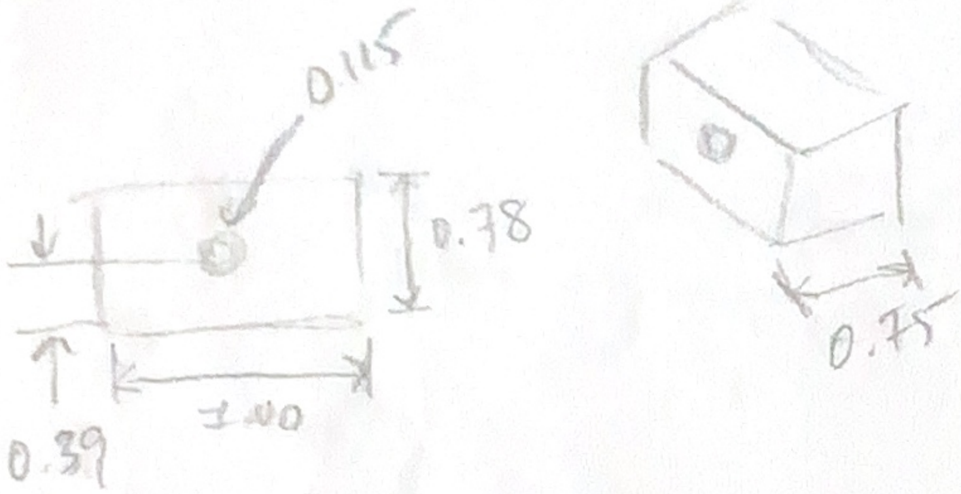
- can there be feedback from the motor on how far it's spun?

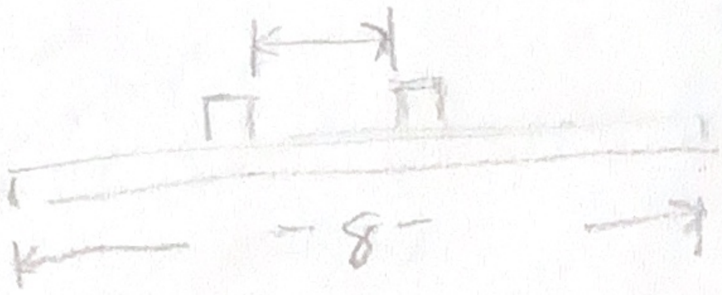
2/10/26

BASE







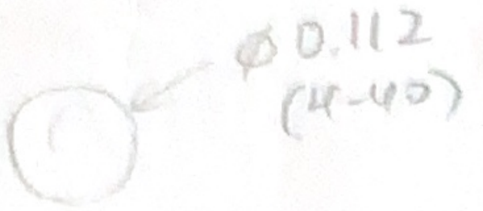


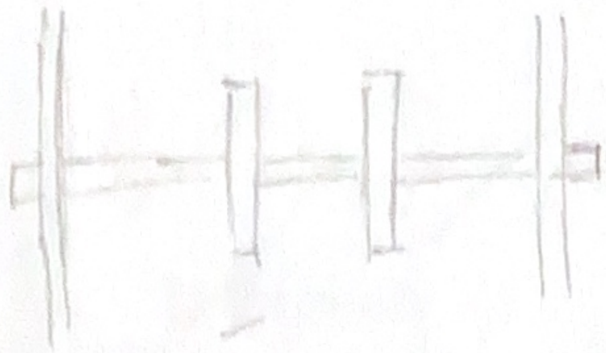
2/20/26

LEAD SCREW

Travel distance per turn -

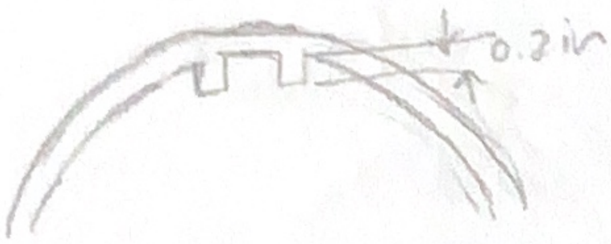
RODS (CONNECTING SCISSOR & CYLINDER)

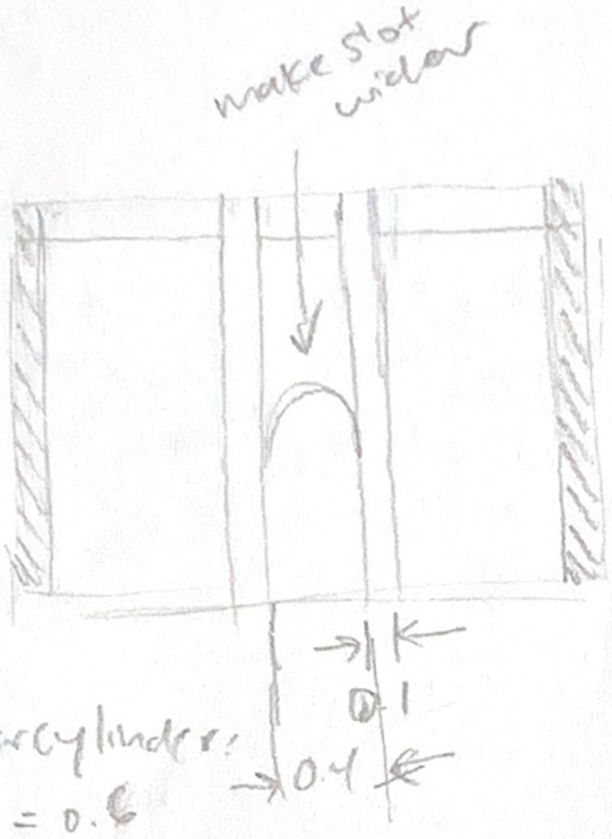




Length: $\emptyset_{CYLINDER} + 0.15$

RAILS FOR SUPPORT





→ print new for 5.7, 5.1, 4.5

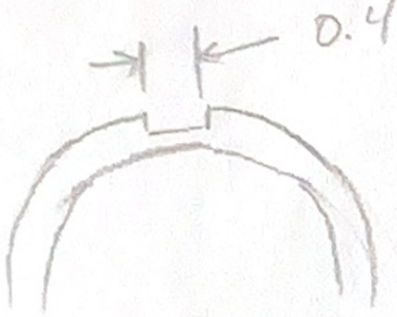
⇒ FOR OLD: glue on rails

→ cut from outside of inner cylinder:

$0.4 + 2(0.10) = 0.6$

↓

0.68



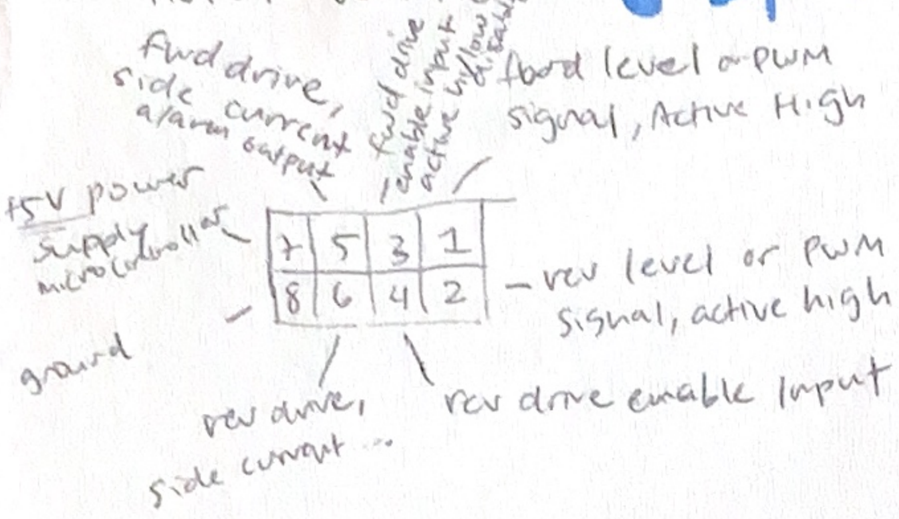
MOTOR CONNECTIONS

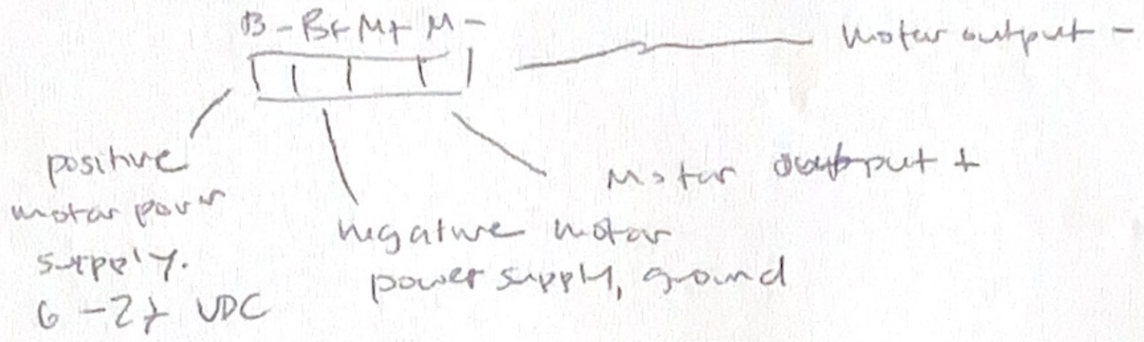
3/11/26

HRS380P-7108
DC 3.6V
2023.04

BTS7960 High Current 43A H-Bridge Motor Driver

- INPUT VOLTAGE : 6-27 VDC





CONTROL INPUT PIN

OUTPUT PIN

MOTOR SPINNING:

- need gear ratio to slow rpm

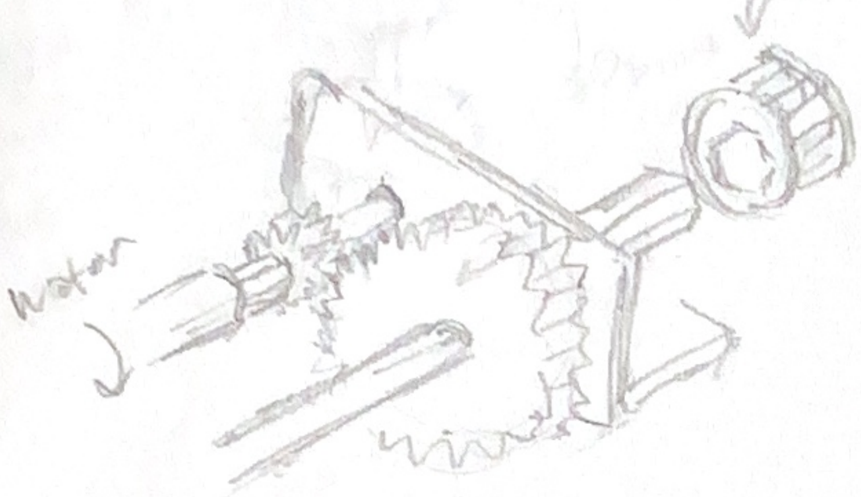
MOTOR GEARBOX FLOWERS

3/4/26

Goal for speed: ~60 RPM

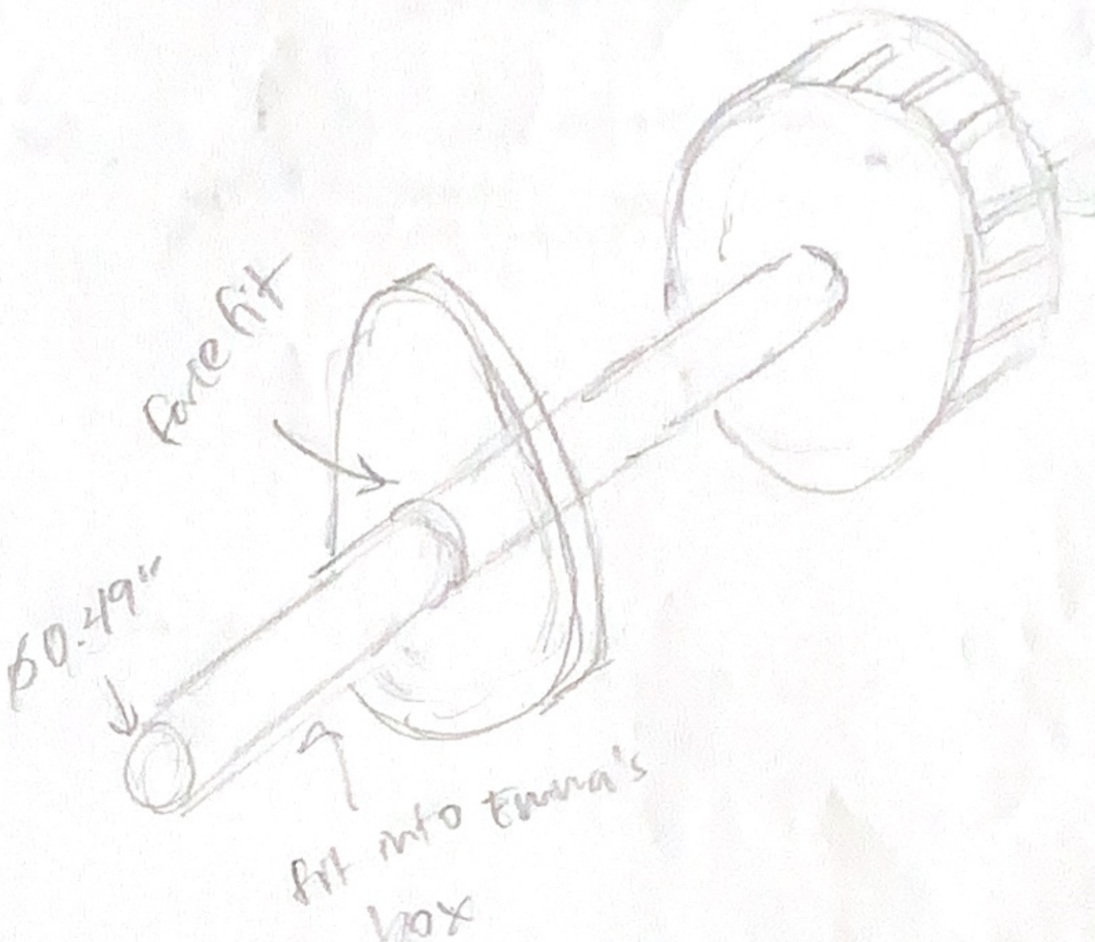
Motor goes 6000 RPM

Current situation:



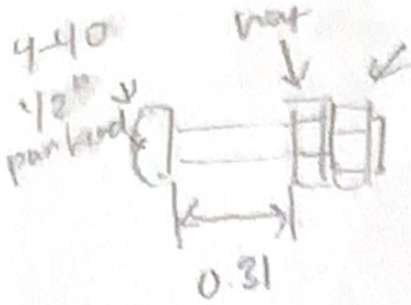
GR: $\frac{48}{16}$

32 DP



SCISSORS - Post printing 1

→ THICKNESSES & BOLTS



⇒ arm thickness = 1.125 → 1.5"

forgot about washer ⇒ no change

→ HOLE TOLERANCE
↳ could be smaller



⇒ hole ϕ = 0.136 → 0.115 → 0.125
↑
too small

→ STABILIZING



one long rod thru each layer

CYLINDERS - Post Printing 1

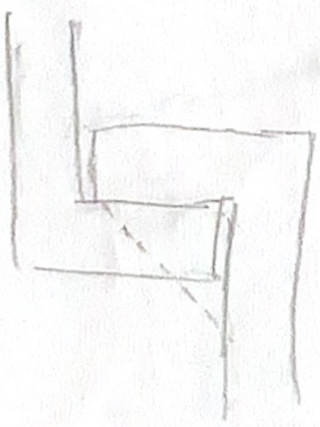
→ WALL THICKNESS
↳ could be smaller

* maybe thicker on the top

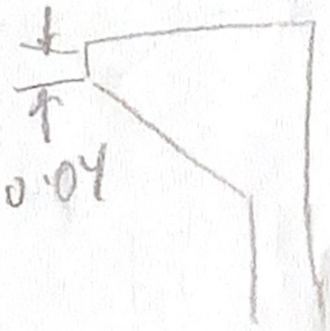
⇒ wall thickness = 0.1" → 0.08"

→ INFILL
↳ could be less
⇒ infill = 05% → 10%

→ LEDGE
↳ maybe fillet so no supports are needed?

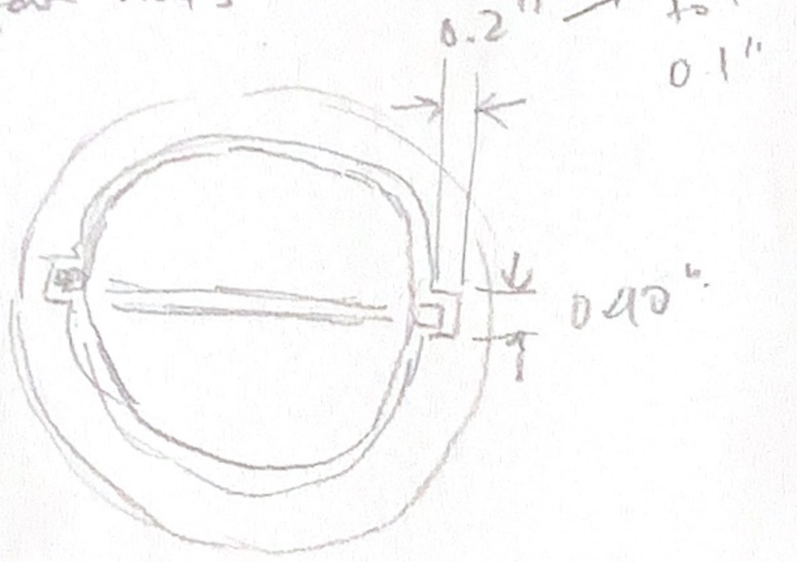


45° chamfer



← make top thinner

↳ add slots for clearance for nuts



CYLINDERS - Post Printing 2

3/15/26

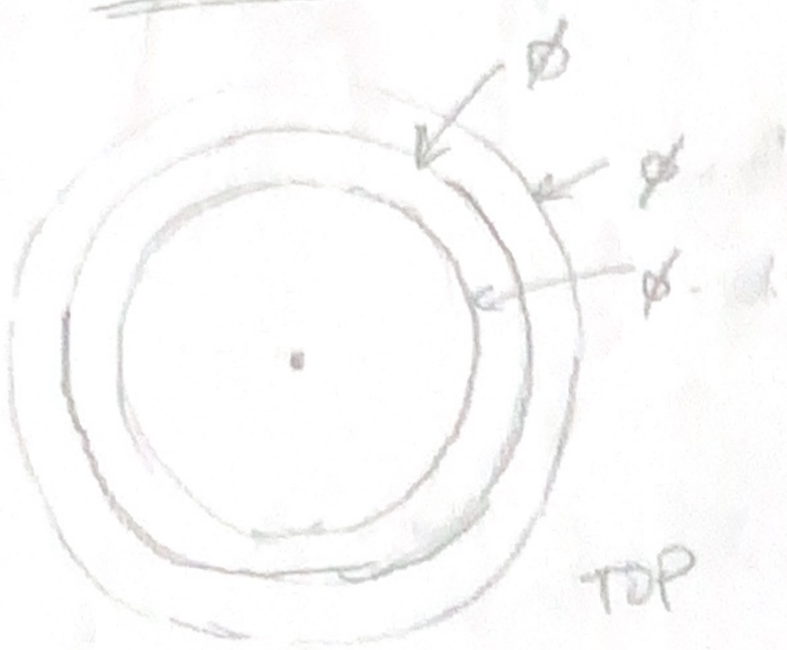
→ TOP CYLINDER
- remove fillers on top ✓
- higher tolerance on top for scissors
- ~~try to trigger another~~

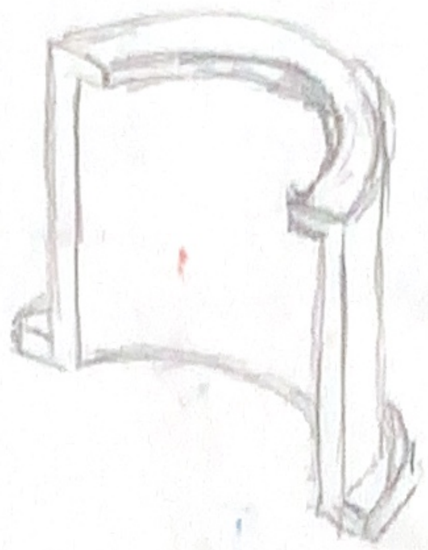
→ TO PRINT
- ~~new length + half arrows~~

→ OTHER TO THINK ABOUT
- better assembly for latches on big holes - first tape on
- how to secure ends? (straight edges?)

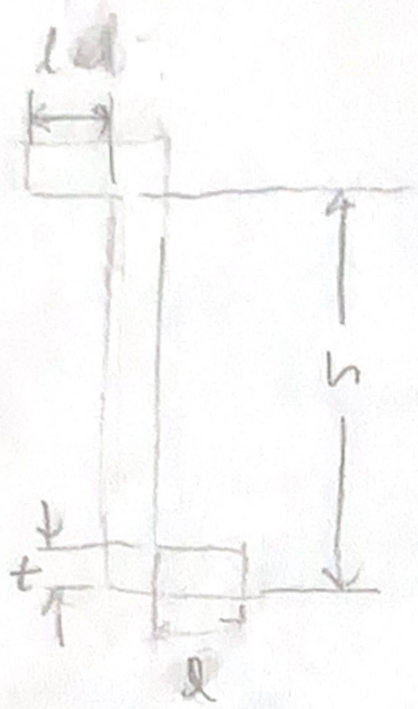
1/11/26

CYLINDERS





n	Diameter	hole
5	4.5	4.375
4	5.1 (5.4)	3.575
3	5.7 (6.0)	2.775
2	6.3 (6.6)	1.975
1	6.9 (7.2)	1.175
0	7.5 (8)	



CROSS SECTION

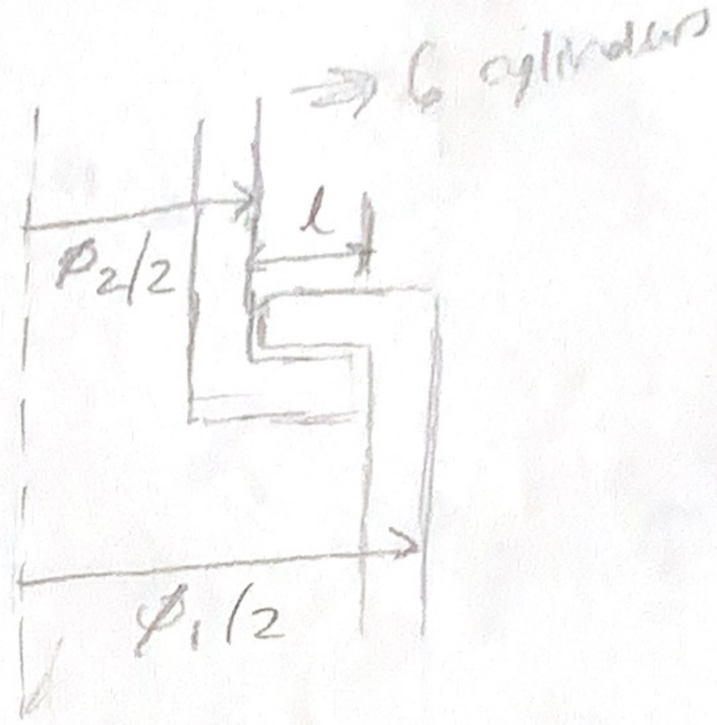
Thickness (t) : 0.08

Height (h) : 9.75

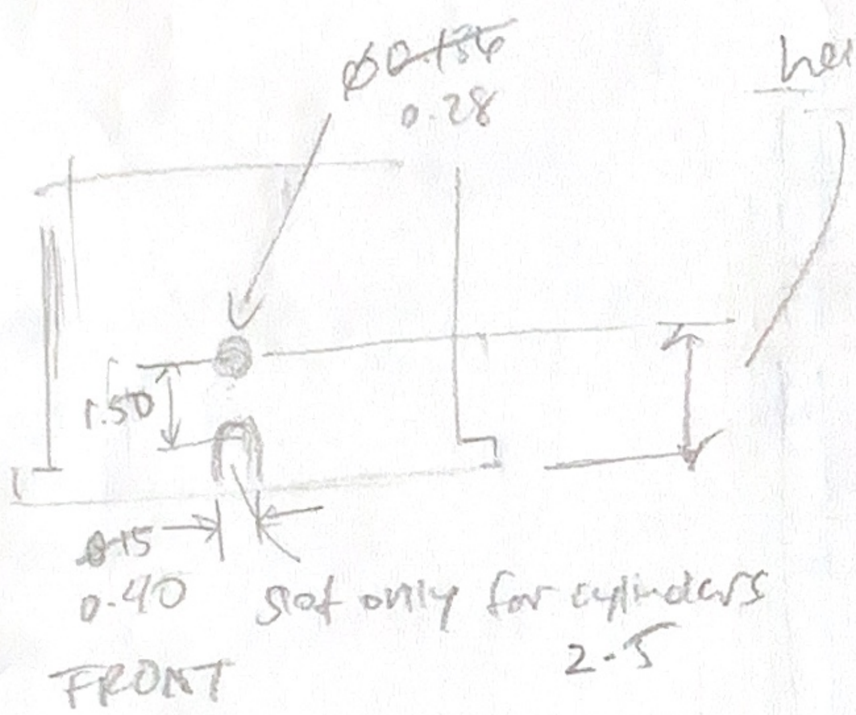
radius Difference (d) : 0.8

Tolerance : 0.008

Ledge (l) : a - t - tol

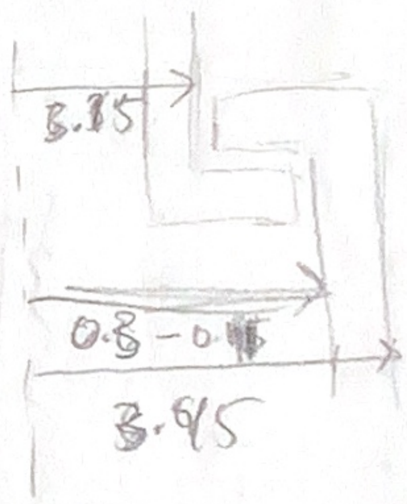


for top :
4.85 (for clearance)

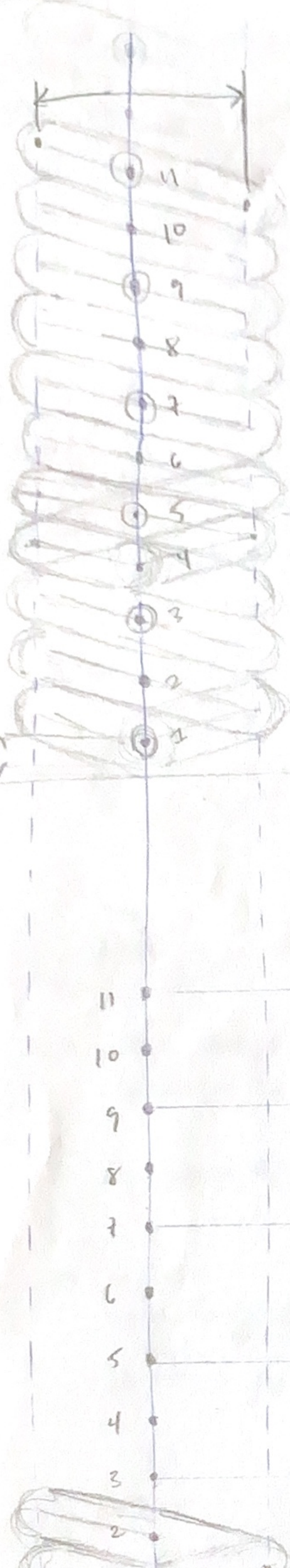


height of hole for cylinder n:

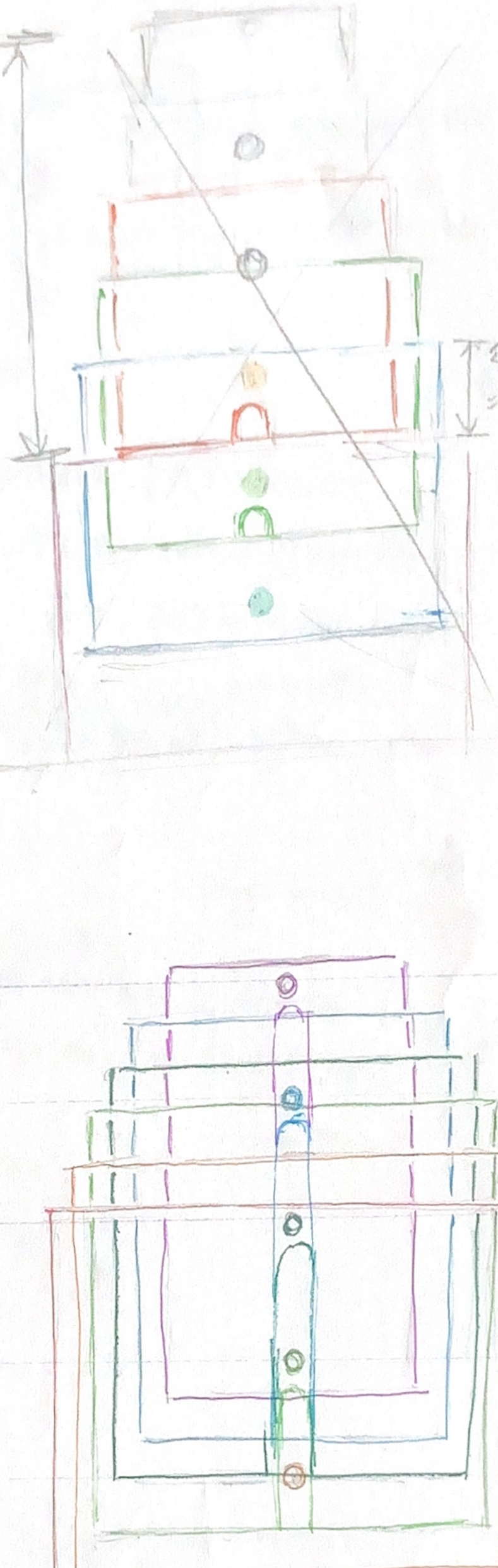
0.375 + 2n(0.36) - n(0.72)

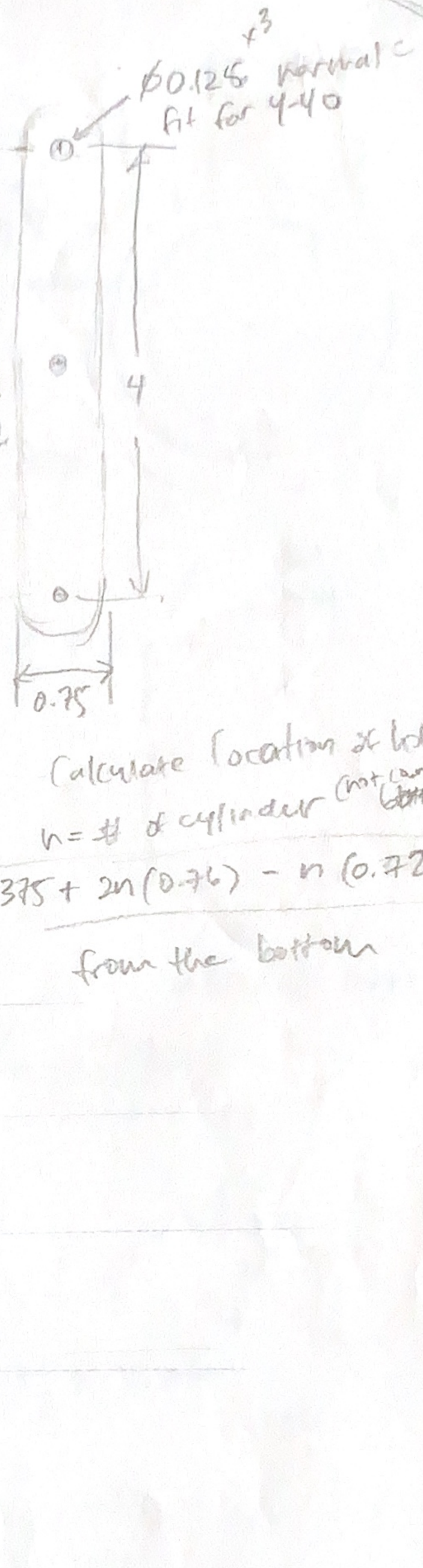


SCISSORS 11 NODES → 7.60 btw 1st and last nodes



8.35
-4.75
=3.6



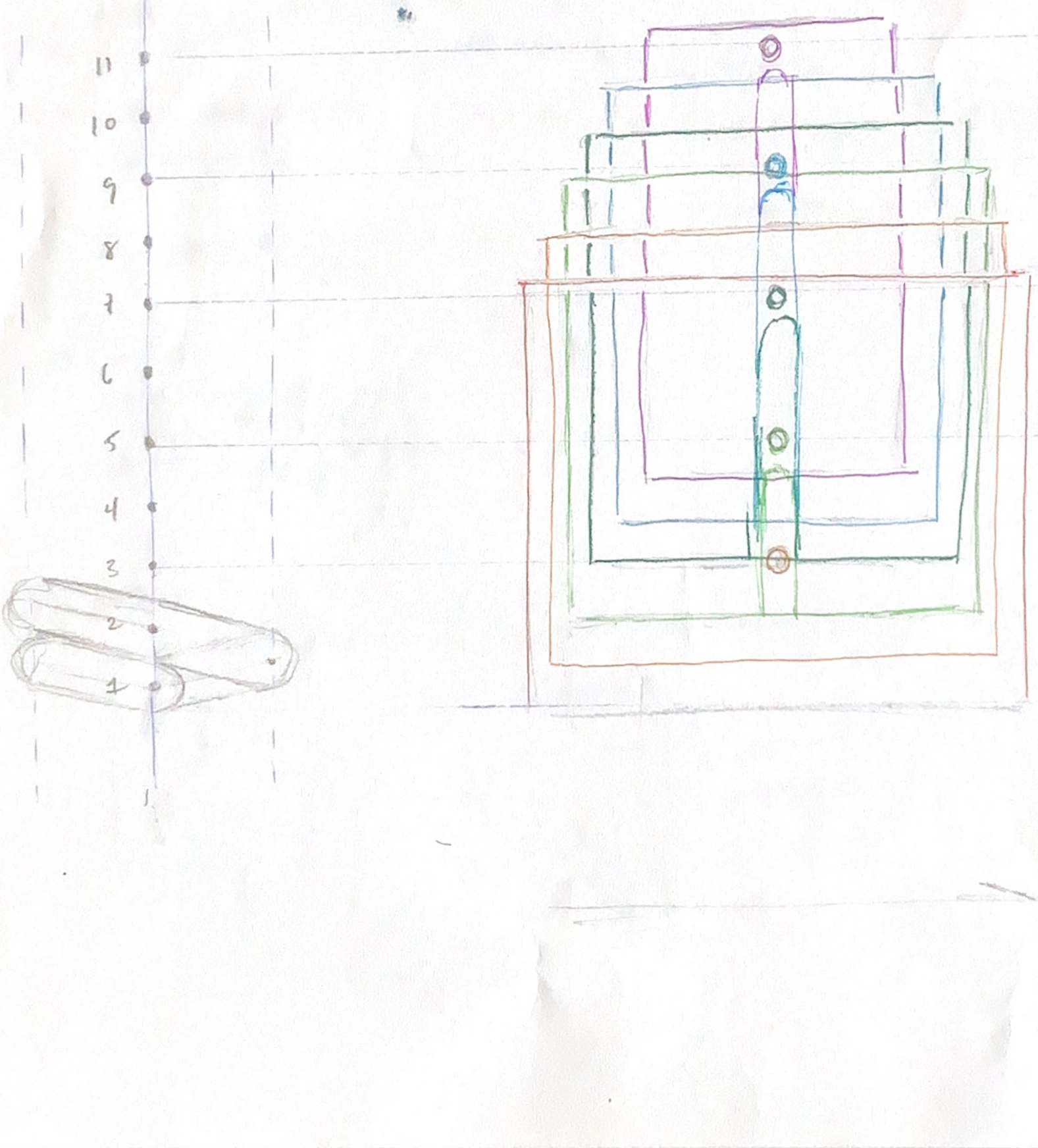


Calculate location of hole

n = # of cylinder (not counting bottom)

$0.375 + 2n(0.76) - n(0.72)$

from the bottom

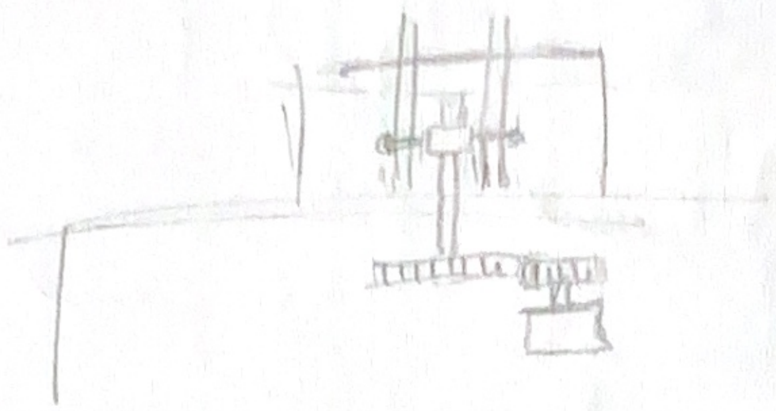


LEAD SCREW FOR TRUNK

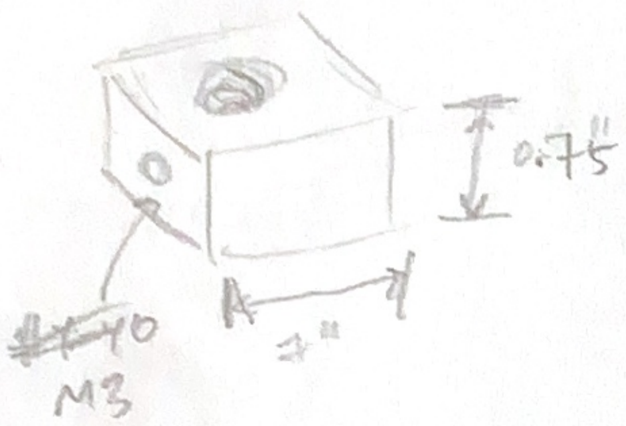
2/18/26

Diameter. M10

Lead: 2mm/rev



NUT.



→ HEATSET HOLE

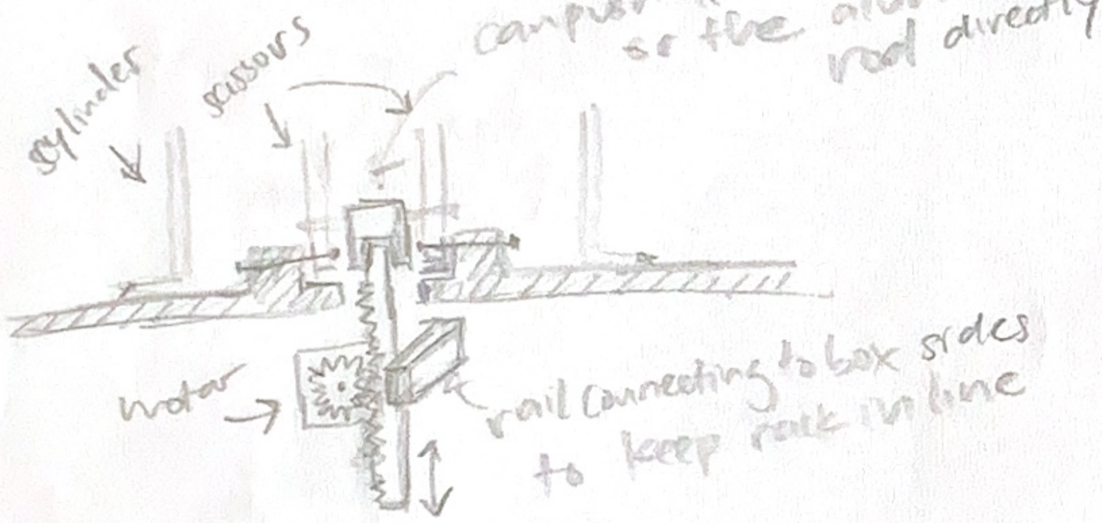
Ø 0.16141732"

depth 0.18"

notes:

- bake along layer lines from ~~pattern~~ chair

Other actuation methods?



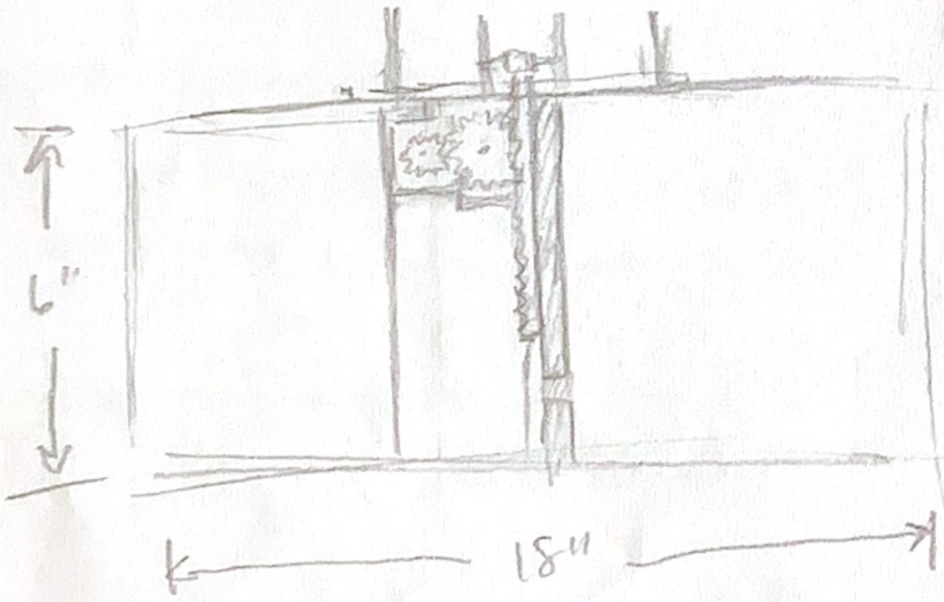


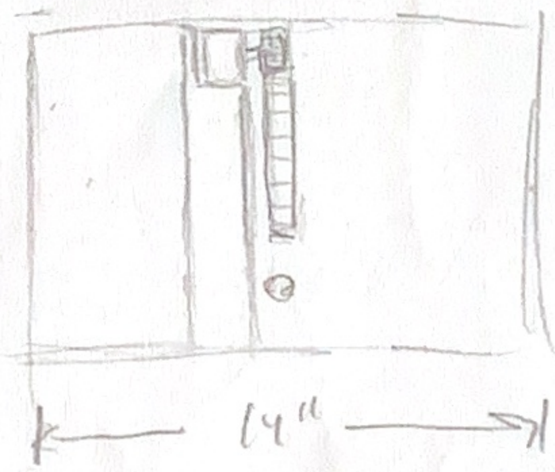
0.84"



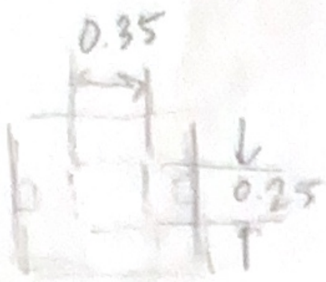
4"

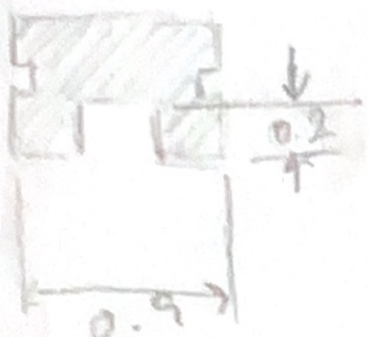
→ rack = 3" + 1" for the length of the motor



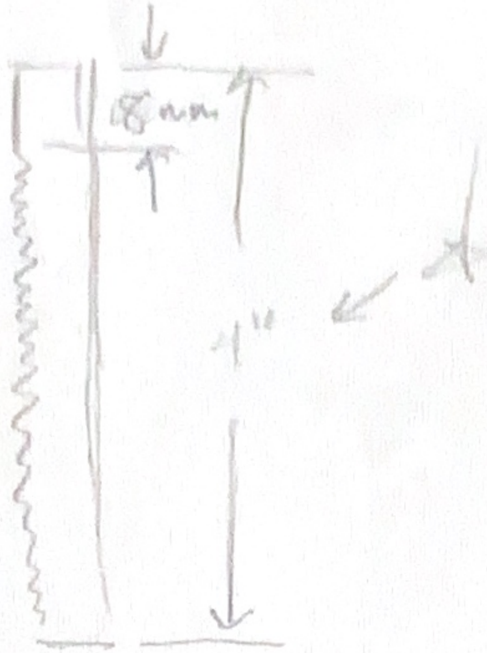


RACK - SCISSOR CONNECTOR

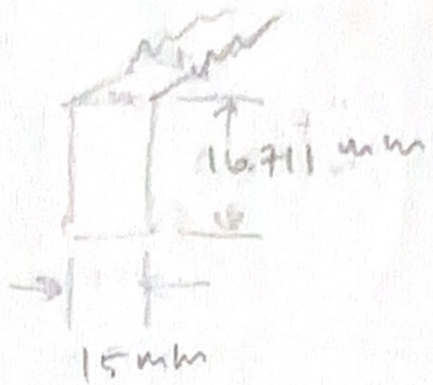




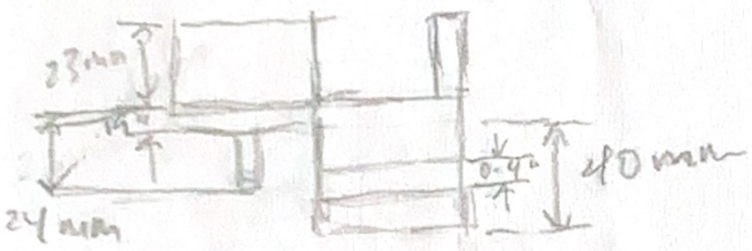
RACK



MODULE : 1.5
20° pressure angle



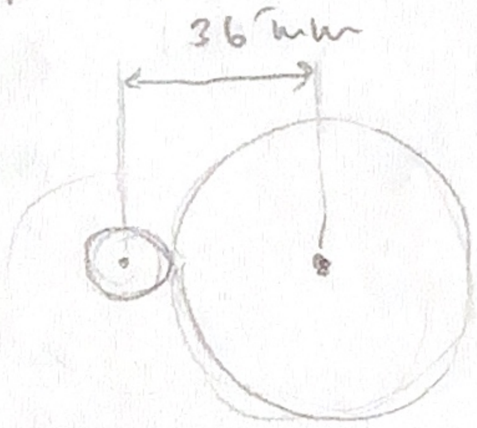
MOTOR SUPPORT



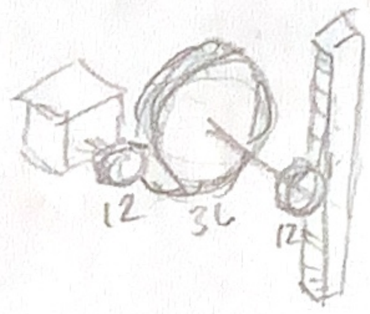
make gears less thick

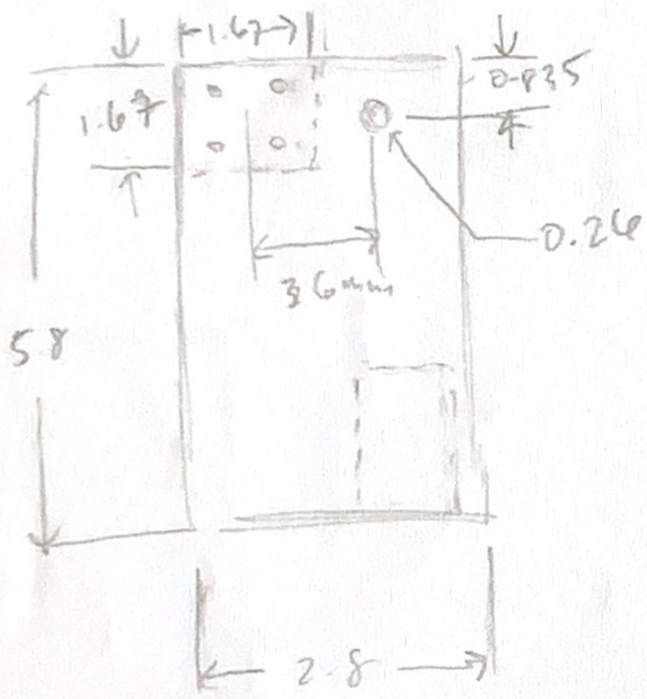
GEARS

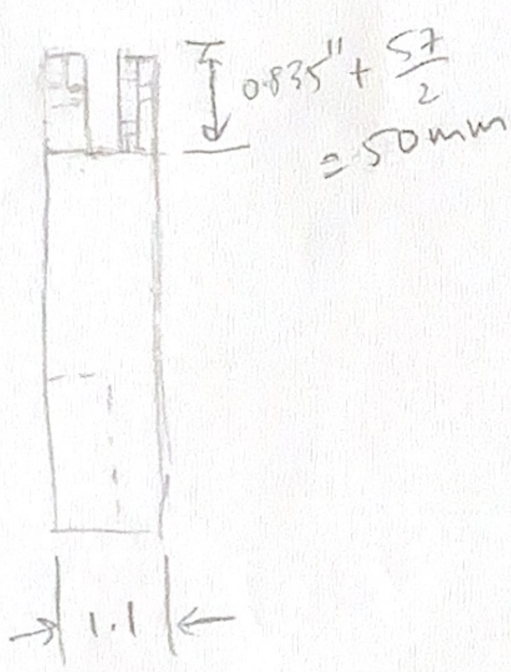
motor
GR: 36:12



PINION: 12T

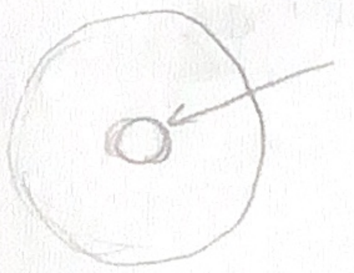






PD	18	54	mm
OD	21	57	mm

$36 + \frac{21}{2} + \frac{57}{2} = 75$ mm

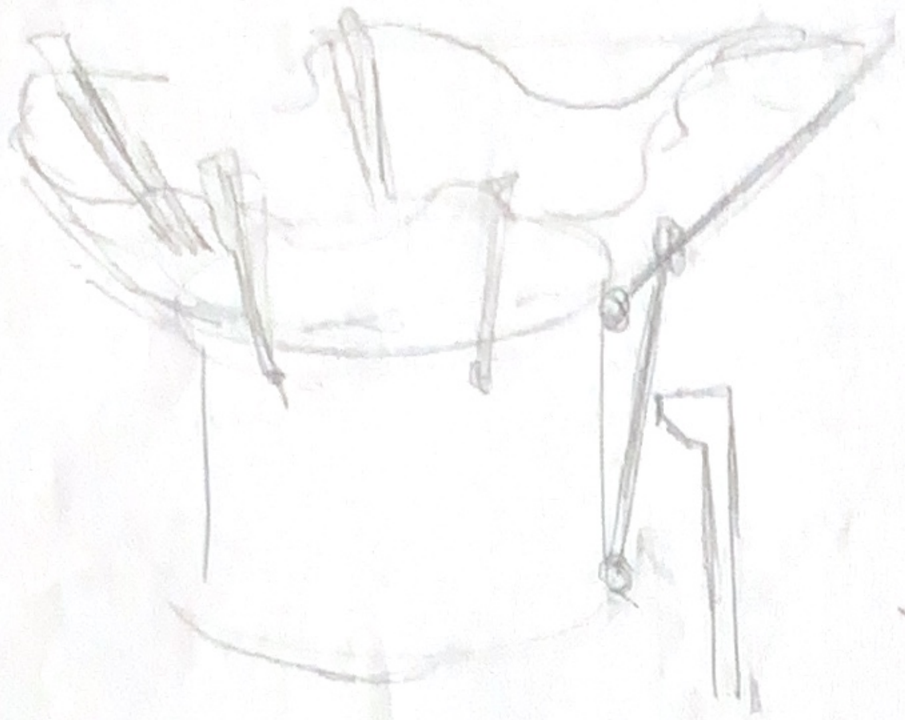


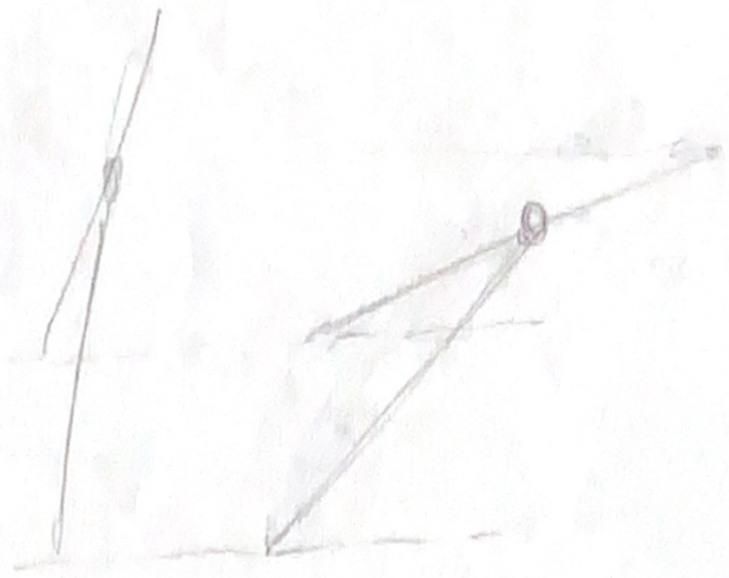
INITIAL PRINTING NOTES - 4/24/22

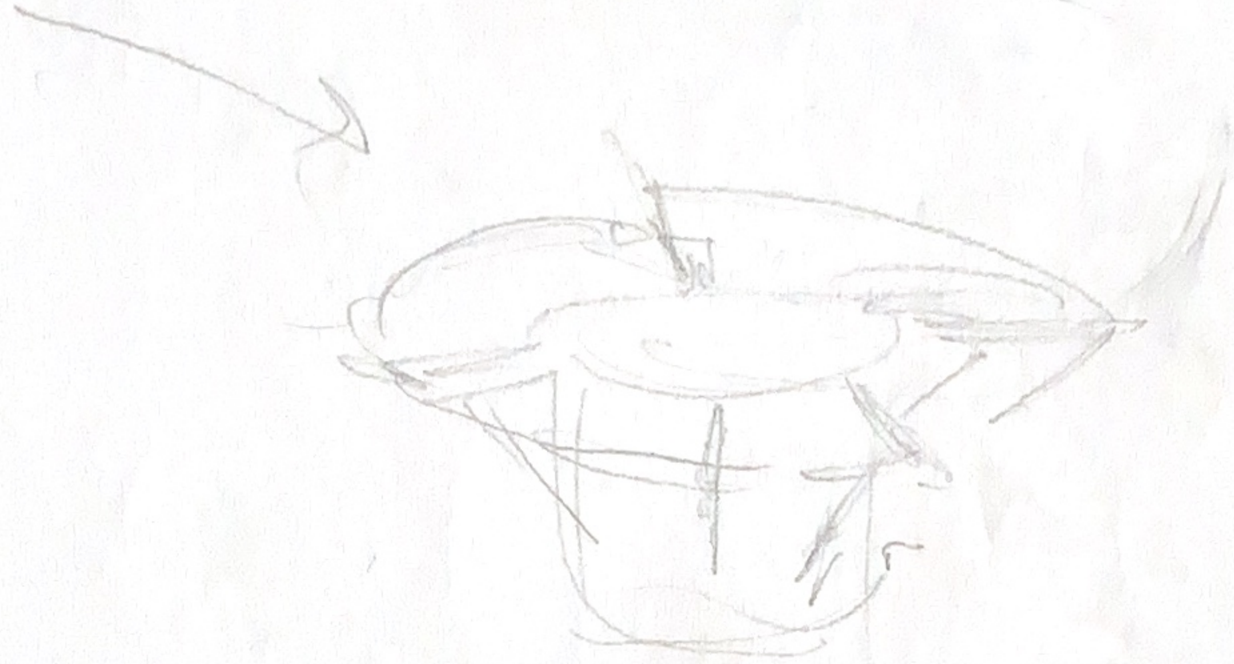
- teeth too small - unstable
- tighter tolerance btw rack & scissor connect
- bigger clearance on gear shaft holes

BRANCHES

2/15/26





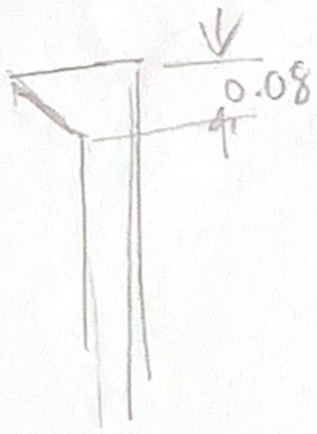


CHANGES TO CYLINDERS

→ need more clearance btw top cylinder and second to top

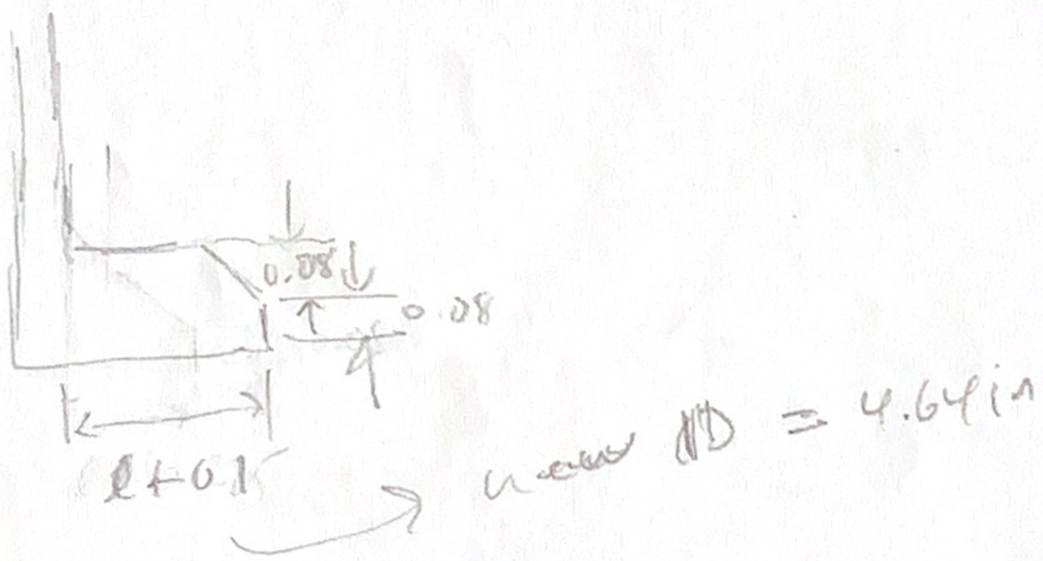
1"? ⇒ Add 0.5" to all ϕ besides top

⇒ SECOND TO TOP

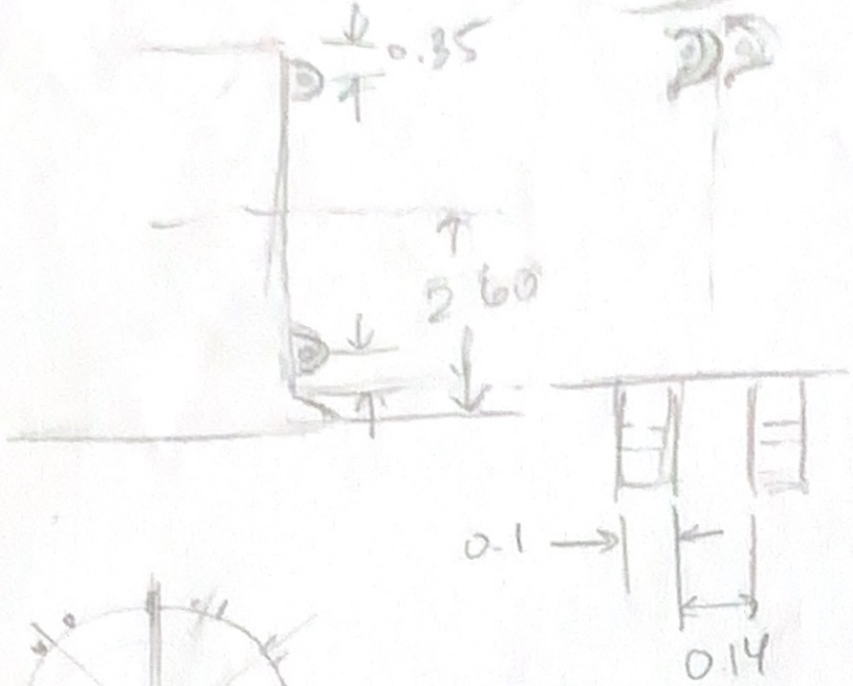


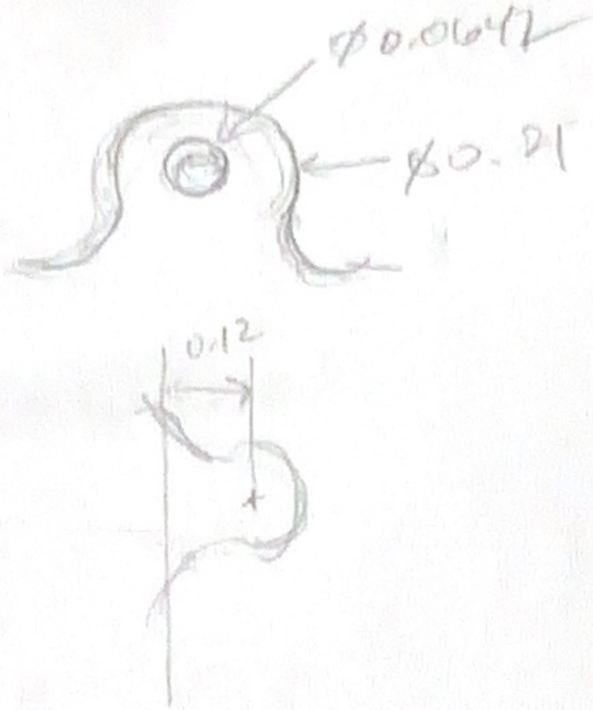
⇒ TOP

2/20/26

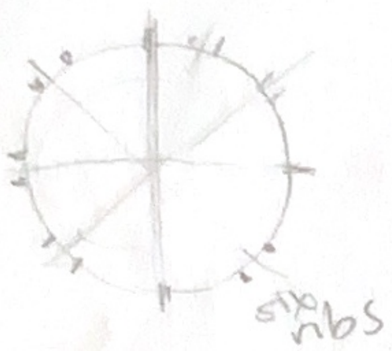


→ TOP CYLINDER

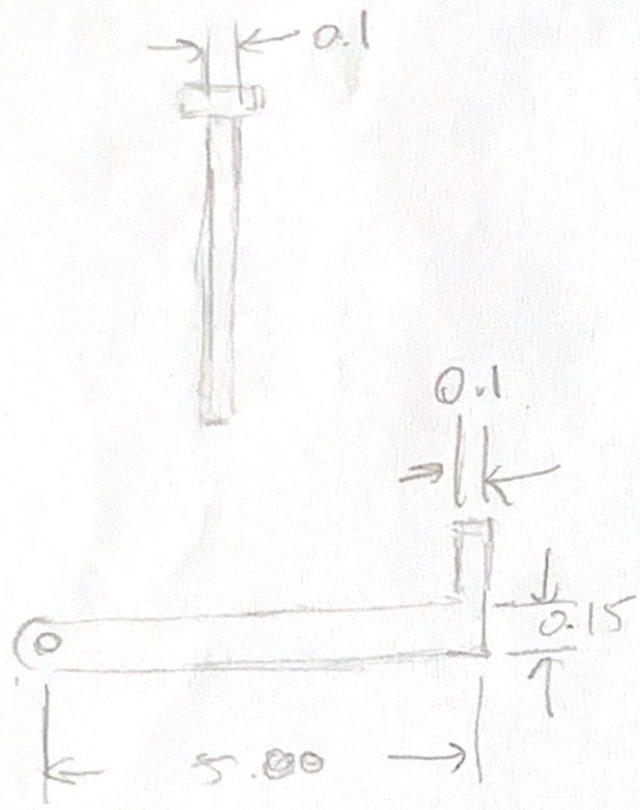




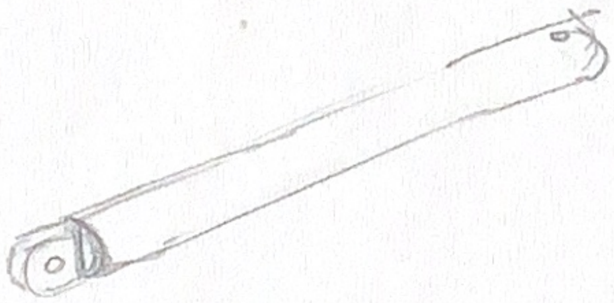
Holes: Ø 0.0647

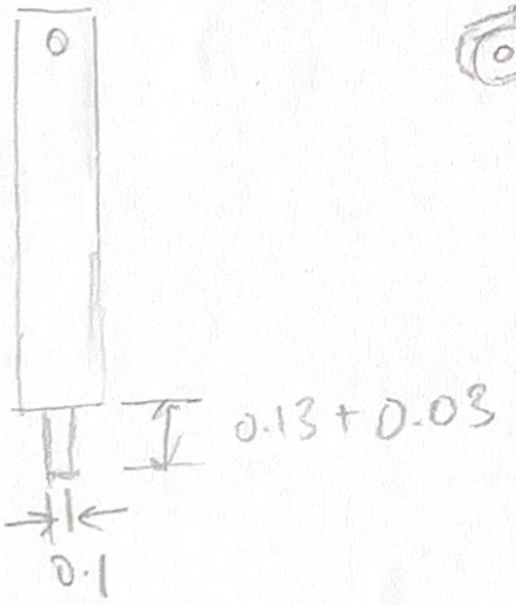


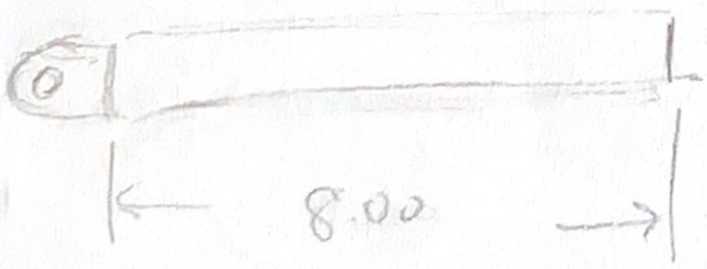
→ RIBS

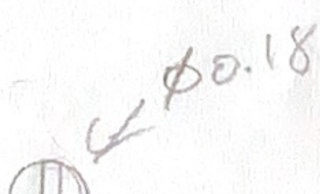












top